

Entropy Collapse Has a Direction: Correcting the Visitation Channel in Token-Level Entropy Control for RLVR

Anonymous ACL submission

Abstract

Reinforcement learning with verifiable rewards (RLVR) reliably drives a policy’s entropy toward zero, and a growing family of methods intervenes by re-weighting the policy loss with a per-token estimate of how much each token’s update costs in entropy. We show that every such method estimates only half of the relevant quantity. The gradient of the *trajectory* entropy splits exactly into two terms: a *local* channel, which asks how the conditional distribution at a visited state flattens or sharpens, and a *visitation* channel, which asks how the probability of *reaching* that state changes. Existing token-level estimators capture the first and are structurally blind to the second. We show the blindness is worse than an omission: the local estimator is exactly zero at a uniform branch point, that zero is the correct first-order answer, and the visitation term’s prefactor is simultaneously maximal there — the two channels are *anti-correlated*, so no refinement of the local term can recover the missing signal. We then derive an estimator for the visitation channel. The reward-to-go implied by the decomposition is the remaining conditional entropy H_{togo} , which we forecast with parallel multi-token-prediction heads at the cost of one extra forward pass — though we also measure that forecast to be close to a scalar multiple of a quantity the trainer already computes, and report the head-free control that follows; the sibling-prefix baseline that keeps the estimator unbiased is supplied by a tree-structured rollout in which several samples share a prefix and diverge at scheduled depths. The resulting advantage A_H sums to zero within a sibling set by construction, so the method cannot change the average amount of damping — it can only decide which of two siblings is damped. We give the full derivation, the four approximations it requires and what each discards, and a five-arm design that isolates the contribution of the forecast’s *value* from that of the apparatus carrying it.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model’s output distribution. Measured as the mean per-token entropy of the policy, this sharpening is fast, monotone, and, past a point, coincident with the end of useful learning. The standard reading is that the collapse is the problem, and the standard remedy is to spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose the most entropy, or — in the most recent line of work — a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token’s update destroys.

We argue that the accounting behind that remedy is incomplete, and that the incompleteness is structural rather than a matter of estimator quality.

Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution the policy induces over whole responses to a prompt x . The autoregressive factorization turns it into an expectation of a sum of per-token conditional entropies, and the gradient of that expression splits, by the product rule, into exactly two terms. The first holds the visited states fixed and asks how the conditional distribution at each one changes: this is what a per-token entropy estimate measures. The second holds the per-state entropies fixed and asks how the probability of *visiting* each state changes. An update that concentrates mass on one continuation makes the states downstream of that continuation more likely and the states downstream of its abandoned siblings less likely; the diversity those siblings carried leaves the trajectory distribution even though no conditional distribution anywhere became more peaked.

Our first contribution is to show that the local channel is not merely blind to the second term but *anti-correlated* with it. The local estimator factors into the advantage, a saturation factor, and the deviation of the sampled token’s surprisal from the

mean surprisal — and that last factor has mean zero by the definition of entropy. At a branch point where several continuations are close to equiprobable it vanishes identically. The vanishing is not an estimator failure: a uniform distribution maximizes entropy, so the first-order change is genuinely zero and the damage is second order (Section 4.4). Meanwhile the visitation term’s prefactor is $1 - \pi_\theta(y_t | s_t)$, which is maximal precisely at those flat states. The local channel therefore spends its budget where the conditional entropy is fragile — rare tokens sampled from confident distributions — and spends nothing where the trajectory entropy is most at risk.

Our second contribution is an estimator for the missing channel that requires no additional rollouts. Eliminating the non-causal cross terms turns the visitation term into a REINFORCE estimator whose reward-to-go is the sum of the *remaining* conditional entropies, H_{togo} ; subtracting any function of the state leaves it unbiased. We estimate H_{togo} with parallel multi-token-prediction (MTP) heads read off the hidden state the policy already computes, and we obtain the state-only baseline from rollouts that share a prefix. Plain group-relative rollouts almost never share prefixes, so we schedule the sharing: a tree-structured sampler branches a single root at fixed depths, and the resulting sibling sets give the baseline an actual support (Section 9).

The resulting per-token score A_H is a deviation from a sibling mean and therefore sums to zero over each sibling set. This has a consequence we state up front because it determines what evidence can count: the method *cannot* raise the average weight at branch points, only spread the weights of siblings apart. “Total entropy went up” is therefore not a prediction of this method, and an experiment looking for it is testing a different hypothesis (Section 9.3).

We make the following contributions.

- An exact two-channel decomposition of the trajectory-entropy gradient, and an argument that the channel captured by current token-level entropy-control methods is anti-correlated with the channel they miss (Sections 4–4.4).
- A derivation of the missing channel into a per-token advantage A_H with no approximation up to the point of estimating H_{togo} , together

with an explicit ledger of the approximations that estimation requires, the sign of each one’s error, and the mechanism that absorbs it (Sections 5–10).

- A tree-structured rollout that supplies the sibling-prefix baseline, and a bound showing what fraction of positions can carry a non-zero A_H under *any* sampler (Section 9).
- A five-arm experimental design that separates the contribution of the forecast’s value from the contribution of the apparatus that delivers it, by running the full pipeline twice with the forecast’s value destroyed in two different ways (Section 11).

2 Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented, and a range of interventions add an explicit entropy term to the objective or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization does not prevent the collapse and that continuing to train under it degrades accuracy (Anonymous, 2025), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Assumption. Both Hao et al. (2026) and this work rely on one structural assumption, which we restate so that our approximation ledger can refer to it.

Assumption 1 (parameter-independent softmax; Hao et al., 2026). For any state s , each action a carries an independent logit $z_{s,a}(\theta)$, so that a gradient step on the sampled token does not substantially move the logits of the other tokens at that state.

It is false for a network with shared parameters. We inherit it rather than introduce it, but our construction adds a second estimator that rests on it.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token’s update induces (Hao et al., 2026). That work observes that earlier interventions are “indirect” and that their effect is therefore limited and

can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Wang et al. (2025d) show that restricting policy-gradient updates to a small minority of high-entropy “forking” tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Local stochasticity without global diversity. Wu et al. (2025a) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Multi-token prediction. We use Medusa-style parallel prediction heads (Cai et al., 2024) not for their intended purpose (speculative decoding) but as an entropy forecaster: we need the entropy of the distribution over a token k steps ahead, not a good sample of that token. Section 6 shows that this choice of estimator has a signed, monotone error which we characterize and correct.

3 Preliminaries

Notation. A prompt is x , a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot | s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta | x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a | s_t)$, so that $H = \mathbb{E}_{a \sim \pi_\theta}[I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \text{mean}_j r_j) / \text{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function

with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and instead multiply each token’s contribution to the clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from a first-order estimate of the entropy change that token’s update induces. Their Theorem 1 gives that estimate in closed form: under a parameter-independent softmax (their Assumption 1; our 2), the change in the conditional entropy at a *fixed* state s after one GRPO step is, to first order,

$$\Omega(s) = -\frac{\eta}{L} \mathbb{E}_a \left[\frac{\mathbb{I}_{\text{clip}}(s, a) A(s, a)}{\pi_{\text{old}}(a | s)} \cdot \pi_\theta(a | s) (1 - \pi_\theta(a | s)) \cdot (\log \pi_\theta(a | s) + H) \right], \quad (1)$$

where η is the learning rate, L the total decoded length in the update, and $\mathbb{I}_{\text{clip}}$ the PPO clipping indicator. The implementation evaluates the bracket at the sampled token rather than in expectation. The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, with

$$\alpha_u = \exp\left(-\alpha \frac{|\Omega_u|}{\max_v |\Omega_v|}\right), \quad \lambda_{\min} = e^{-\alpha}, \quad (2)$$

a batch-normalized exponential map into $[\lambda_{\min}, 1]$, so the weight can only attenuate. The factor $1/\pi_{\text{old}}$ in Eq. (1) will matter later: it makes Ω heavy-tailed, which is why the two channels must be normalized before they are combined (Section 8).

4 Two Channels

4.1 Step 1: the chain rule

The object we would like to protect is the entropy of the distribution the policy induces over *whole responses*,

$$\mathcal{H}(\pi_\theta | x) := -\sum_y \pi_\theta(y | x) \log \pi_\theta(y | x), \quad (3)$$

a sum over an exponentially large set. The autoregressive factorization $\pi_\theta(y | x) = \prod_t \pi_\theta(y_t | s_t)$ turns the logarithm into a sum and lets us exchange

the order of summation:

$$\begin{aligned}\mathcal{H} &= - \sum_y \pi_\theta(y | x) \sum_{t=1}^T \log \pi_\theta(y_t | s_t) \\ &= \sum_{t=1}^T \mathbb{E}_{y \sim \pi_\theta} [-\log \pi_\theta(y_t | s_t)].\end{aligned}\quad (4)$$

Each summand is still an expectation over whole responses, but the integrand depends on y only through $y_{\leq t}$. Conditioning on the prefix and applying the tower rule replaces the inner expectation with a per-position entropy,

$$\begin{aligned}\mathbb{E}_y [-\log \pi_\theta(y_t | s_t)] \\ = \mathbb{E}_{y_{<t}} \left[\underbrace{\mathbb{E}_{y_t \sim \pi_\theta(\cdot | s_t)} [-\log \pi_\theta(y_t | s_t)]}_{= H(\pi_\theta(\cdot | s_t))} \right],\end{aligned}$$

because the inner expectation is, by definition, the entropy of the conditional distribution at s_t . Collecting the terms back under a single expectation gives

$$\mathcal{H}(\pi_\theta | x) = \mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T H(\pi_\theta(\cdot | s_t)) \right]. \quad (5)$$

Equation (5) is an identity, not an approximation. Its left-hand side is what we want to protect; its right-hand side is what instrumentation can measure, since $H(\pi_\theta(\cdot | s_t))$ is read directly off the logits at each position.

The entire argument of this paper follows from one feature of Eq. (5) that is easy to pass over: *the measure $\mathbb{E}_{y \sim \pi_\theta}$ depends on θ as well*. The right-hand side is not a sum of θ -dependent terms over a fixed set of positions — it is a θ -dependent average of them over a θ -dependent set of trajectories.

4.2 Step 2: the product rule

Write the per-trajectory total

$$f_\theta(y) := \sum_{t=1}^T H(\pi_\theta(\cdot | s_t(y))), \quad (6)$$

so that Eq. (5) reads $\mathcal{H} = \sum_y \pi_\theta(y | x) f_\theta(y)$. Both factors of every summand carry θ , so differentiating requires the product rule:

$$\begin{aligned}\nabla_\theta \mathcal{H} &= \sum_y [\nabla_\theta \pi_\theta(y | x)] f_\theta(y) \\ &\quad + \sum_y \pi_\theta(y | x) \nabla_\theta f_\theta(y).\end{aligned}\quad (7)$$

The second sum is already an expectation. The first is not, because $\nabla_\theta \pi_\theta$ is not a probability distribution; the score-function identity $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ restores it to one. Substituting and naming the two terms,

$$\begin{aligned}\nabla_\theta \mathcal{H} &= \underbrace{\mathbb{E}[\nabla_\theta f_\theta(y)]}_{\text{(L) local}} \\ &\quad + \underbrace{\mathbb{E}[f_\theta(y) \nabla_\theta \log \pi_\theta(y | x)]}_{\text{(V) visitation}}.\end{aligned}\quad (8)$$

Equations (3)–(8) contain no approximation of any kind. The split into two channels is forced by the product rule, not chosen.

4.3 Reading the two terms

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the subtree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_\theta \mathcal{H}$.

4.4 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level.

Collecting the constants of Eq. (1) into $c := \eta \mathbb{I}_{\text{clip}} A / (L \pi_{\text{old}})$ and rewriting in terms of surprisal gives, for the contribution of a single action a ,

$$\Omega \propto c \pi_a (1 - \pi_a) (H - I_a), \quad (9)$$

and $H = \mathbb{E}_{a \sim \pi} [I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the

magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy. Nothing in c depends on the shape of $\pi(\cdot | s)$, so the argument below is unaffected by the clipping indicator or the importance weight.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a , so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability — are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer. This is the step that closes the argument, and it follows from Hao et al.’s own derivation. Differentiating the conditional entropy with respect to a single logit gives (Hao et al., 2026, Eq. 20)

$$\frac{\partial H(\pi_\theta | s)}{\partial z_{s,a}} = -\pi_\theta(a | s)(\log \pi_\theta(a | s) + H(\pi_\theta | s)), \quad (10)$$

and at a uniform distribution $\log \pi_\theta(a | s) = -\log k = -H$ makes the bracket vanish for every a . So $\partial H / \partial z_a = 0$ identically: Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^\delta + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^\delta}{S}, \quad H''(0) = -\frac{k-1}{k^2}, \quad (11)$$

with $H'(\delta)$ of sign $-\text{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4} , roughly two orders of magnitude below that of a peaked distribution under the same perturbation. As far as local entropy is concerned, uniform branch points are genuinely safe, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are anti-correlated. The visitation term’s prefactor, derived in Section 7, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates

distribution	H	$P(\text{split})$	$\mathbb{E} \Omega / A $	$\mathbb{E}[1-\pi_a]$
$q=.99$	0.10	0.08	0.092	0.020
$q=.9, \text{ tail}$	1.25	0.57	1.129	0.190
$q=.6, m=3$	1.11	0.98	0.457	0.587
unif. $k=10$	2.30	1.00	0.000	0.900
unif. $k=100$	4.61	1.00	0.000	0.990

Table 1: The local channel and the visitation channel are ordered oppositely. Distribution family: top token has mass q , the remaining $1 - q$ is spread uniformly over m tokens. $P(\text{split})$ is the probability that $n=8$ samples do not all agree. $\mathbb{E}|\Omega|$ is taken over the token actually sampled at that position.

between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

The statement we can therefore make is stronger than “the local term is incomplete”: *the local term is anti-correlated with the term it omits*, and the omission cannot be fixed within the local channel.

5 Estimating the Visitation Channel

5.1 Causal elimination gives a reward-to-go

The visitation term of Eq. (8) is not yet in a usable form: the factor $f_\theta(y)$ multiplying the score is a property of the *whole* trajectory, so a naive estimator would credit every token with entropy that appeared before it. Causality removes those terms exactly.

Expanding both factors — f_θ by Eq. (6) and the score by $\nabla_\theta \log \pi_\theta(y | x) = \sum_u \nabla_\theta \log \pi_\theta(y_u | s_u)$ — turns (V) into a double sum:

$$(V) = \sum_t \sum_u \mathbb{E}[H(\pi_\theta(\cdot | s_t)) \cdot \nabla_\theta \log \pi_\theta(y_u | s_u)]. \quad (12)$$

Consider a term with $u \geq t$. The entropy $H(\pi_\theta(\cdot | s_t))$ depends on y only through $y_{<t}$, and $y_{<t} \subseteq y_{<u}$, so it is measurable with respect to $y_{<u}$ and may be pulled out of a conditional expectation given $y_{<u}$. What remains is the expected score at a single position, which vanishes for any distribution:

$$\begin{aligned} & \mathbb{E}[\nabla_\theta \log \pi_\theta(y_u | s_u) | y_{<u}] \\ &= \sum_a \pi_\theta(a | s_u) \frac{\nabla_\theta \pi_\theta(a | s_u)}{\pi_\theta(a | s_u)} \\ &= \nabla_\theta \sum_a \pi_\theta(a | s_u) = \nabla_\theta 1 = 0. \end{aligned} \quad (13)$$

By the tower rule every (t, u) term with $u \geq t$ is therefore *exactly* zero — not small, zero. This is the same identity that licenses a baseline in policy gradients, used here to enforce causality rather than to reduce variance.

Only $u < t$ survives. Swapping the order of summation collects, for each token u , the entropies at all later positions:

$$(V) = \mathbb{E} \left[\sum_u \nabla_{\theta} \log \pi_{\theta}(y_u | s_u) H_{\text{togo}}(s_u \oplus y_u) \right], \quad (14)$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_{\theta}(\cdot | s_t))$.

We stress that H_{togo} was not posited. Equation (14) is a REINFORCE estimator whose reward is “the conditional entropy remaining after this token”, and H_{togo} is that reward-to-go.

5.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (14) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \quad (15)$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\leq u}$: the rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO’s group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout’s own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

6 Four Approximations

Equation (15) is exact but not computable: H_{togo} sums conditional entropies at states the update has not visited. Three approximations make it computable and a fourth bounds its influence.

6.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\text{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k H(\pi(\cdot | s_{u+k})). \quad (16)$$

The discount serves two purposes: distant terms carry more forecast error (A2) and therefore deserve less trust, and the sum stays bounded independently of κ .

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (15) consumes only the *difference* between siblings. Any bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

6.2 A2: forecasting the unrealized future

Even Eq. (16) requires the future states s_{u+k} . At update time they do not exist: rollouts came from π_{old} , each state has exactly one sampled continuation, and re-rolling is prohibitive. We replace the realized future with a parallel forecast from the current hidden state h_u :

$$H(\pi(\cdot | s_{u+k})) \longrightarrow H(p_{\text{MTP}}^{(k)}(\cdot | h_u)). \quad (17)$$

Head k is a two-layer residual MLP applied to h_u , followed by the policy’s own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy’s own next-token distribution and the forecast starts from a sane value. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

The error of this substitution has a known sign. Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u , the k -th term of Eq. (16) has expectation $H(Y_{u+k} | Y_{u+1:u+k-1}, S_u)$, whereas the MTP head models the *marginal* $p(Y_{u+k} | S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} | S_u) - \mathbb{E}[H(\pi(\cdot | S_{u+k-1}))] \\ = I(Y_{u+k}; Y_{u+1:u+k-1} | S_u) \\ \geq 0. \end{aligned} \quad (18)$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k .

Equation (18) is not merely a bound: it is the reason the marginal is the quantity we want. It is the fan-out of the continuation — how many ways the next k tokens can unfold — rather than the residual uncertainty along the one path that happened to be sampled.

6.3 A3: per-head affine calibration

Equation (18) predicts a per-head, monotone inflation, so we absorb it with a per-head affine correction fitted by least squares against oracle entropies on warm-up rollouts:

$$\hat{H}_k = a_k H(p_{\text{MTP}}^{(k)}(\cdot | h_u)) + b_k. \quad (19)$$

The fitted coefficients confirm the prediction. On our model the first head has $a_1 = 1.03$ — a one-step forecast has no intervening tokens, so the mutual information in Eq. (18) is zero and the approximation is exact — while a_k drops to 0.10–0.19 for $k \geq 2$. Calibration switches the unreliable heads off on its own.

Combining A1–A3 gives the forecast that is actually computed:

$$\hat{H}_{\text{togo}}(s_u \oplus y_u) = \max\left(0, \sum_{k=1}^{\kappa} \gamma_H^k (a_k H_k(h_u) + b_k)\right), \quad (20)$$

the clamp encoding the fact that an entropy cannot be negative. With $\kappa = 2$ and $\gamma_H = 0.7$ this instantiates to $\hat{H}_{\text{togo}} = 0.72 H_1 + 0.06 H_2 + 0.11$: the second head contributes about 8%, so the effective horizon is close to one step. We regard this as a consequence of Eq. (18) rather than a tuning artifact, and note it as a limitation — a forecaster whose error did not grow with k would leave a longer horizon usable.

Alignment. One detail decides whether the signal is a branch score at all. The hidden state at position u is conditioned on s_u and predicts y_u ; it does not know y_u . The quantity we need, $H_{\text{togo}}(s_u \oplus y_u)$, must therefore be read from position $u + 1$ and shifted back by one. Omitting the shift measures the value of the state *before* the branch rather than the value of the branch taken.

7 From Gradient to Token Weights

Equation (15) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u , changes the trajectory entropy.

How far one step moves a logit. We reuse the logit-space update of Hao et al. (2026, Eq. 23) rather than rederiving it. With $w_u = \text{clip}(r_u, 1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}}) A_u$ the coefficient the policy loss actually applies, Assumption 2 makes a single step move only the sampled token’s logit, $\Delta z_{u,a} = \eta w_u \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi_u} = \eta w_u (1 - \pi_u), \quad (21)$$

which vanishes as $\pi_u \rightarrow 1$: a token that already holds the mass cannot be given more. This is the prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_{\theta} \log \pi_u \cdot \Delta \theta = \Delta \log \pi_u$, substituting into Eq. (15) gives the per-token quantity

$$\text{visit}_u := \widehat{\Delta \log \pi_u} \cdot \text{clip}(A_H(s_u, y_u), -c, +c). \quad (22)$$

A4 is the clip, and it is applied to A_H *before* the product rather than after, so that one token’s influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let forecast error be amplified without bound at tokens with large w_u .

Sign quadrants. Equation (22) defines the collapse channel precisely. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi_u}$ reverses both readings.

8 Combining the Channels

The visitation term could be added to Ω before the weight map, but measured on real rollouts that map destroys the signal: Ω carries a factor $1/\pi_{\text{old}}$, so $\max |\Omega|/\text{median}|\Omega|$ reaches 3×10^4 and the band is spent on one or two outliers while the bulk lands on a point. We therefore apply the correction *after* the map, as a bounded adjustment of the weights the base method has already produced.

Let $\mathcal{S} = \{u : \text{visit}_u \neq 0\}$ be the support — by Eq. (24), the branch points — and let $b = \lambda_{\text{max}} - \lambda_{\text{min}}$ be the width of the weight band. For $u \in \mathcal{S}$,

$$\alpha_u^f = \text{clamp}\left(\alpha_u + \lambda b \tanh(\text{visit}_u / \rho), \lambda_{\text{min}}, \lambda_{\text{max}}\right), \quad (23)$$

with $\rho = \text{rms}\{\text{visit}_v : v \in \mathcal{S}\}$; $\alpha_u^f = \alpha_u$ elsewhere.

Three properties matter. The tanh is odd, monotone and bounded, so it preserves the sign of the visitation term, caps one token’s movement at λb , and does not depend on the absolute scale of visit. The RMS is taken over $\mathcal{S}$ alone rather than over the whole response: normalizing by the whole response would shrink the correction in proportion to how sparse branch points are, which is a property of the sampler and not of the signal. And at $\lambda = 0$ the correction vanishes identically, so the pipeline is bitwise identical to the base method — a regression test enforces this.

Because the map runs first and the correction second, the step size η of Eq. (21) enters only through visit_u/ρ , where it cancels. We fix $\eta = 1$ and sweep λ alone.

The support is part of the treatment. $\mathcal{S}$ is defined by $\text{visit}_u \neq 0$ rather than by a threshold, deliberately: it selects both where the correction lands and, through ρ , how large every correction is, so a tighter test would change what is trained rather than what is logged. One consequence is that $\mathcal{S}$ admits float32 round-off — A_H is a deviation from a mean over token-identical siblings, and $\sum_j g_j/m$ does not reproduce g_i bitwise, leaving $|A_H| \sim 6 \times 10^{-8}$ where it is interpretively zero. Those residuals are $\sim 10^{-6}$ relative to a real branch score and cannot move a weight, but they are counted in the support statistics of Section 12.6, which should be read as an upper bound on the branch-point fraction.

9 Tree-Structured Rollouts

9.1 Why plain rollouts cannot support A_H

$\hat{H}_{\text{togo}}$ is a deterministic function of the causal prefix, so two rollouts sharing $y_{\leq u}$ produce identical forecasts. Combined with Eq. (15) this gives a sharp characterization:

$$A_H[i, u] \neq 0 \iff \begin{array}{l} \text{siblings share } y_{<u} \\ \text{but diverge at } y_u, \end{array} \quad (24)$$

that is, u is a branch point of the group. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so $m = 1$ almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths $d_1 < d_2 < d_3$ with branching factor 2, yielding $n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (24) can fire.

9.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not, and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (24) it counts branch points, and n sequences form a trie with at most $n - 1$ internal branching nodes.

Since the one-position alignment shift widens each branch node into two columns, the fraction of positions carrying a non-zero A_H is bounded by $2(n - 1)/T$ for *any* sampler whatsoever. With $n = 8$ and responses of $\sim 10^3$ tokens this ceiling is on the order of a percent. The tree’s role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

9.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \text{sib}(s_u)} A_H(s_u, y_u^{(j)}) = 0. \quad (25)$$

The correction in Eq. (23) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. “Mean entropy at branch points increased” is *not* a prediction of this method and cannot serve as evidence that it works; the corresponding prediction is that the *variance across siblings* widens. A method designed to raise the mean is a different and weaker hypothesis — it does not need a forecast at all, only a support — and we include it below as a control arm.

10 The Approximation Ledger

Table 2 collects every departure from Eq. (8).

The strongest assumptions are A2 and B2. A2 has a known sign and a known monotonicity (Eq. 18), which is what makes the affine correction in A3 an appropriate remedy rather than a curve fit. B2 — that updating one token does not change the conditional distribution at other positions — is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding a second channel adds a second copy of its error.

11 Experimental Setup

11.1 Model, data, and optimization

Within a backbone, all runs fine-tune the same base model on the same data with the same optimizer settings, and differ only in the intervention. Our primary backbone is a 1.5B-parameter mathematics-specialized model; the full five-arm decomposition and every ablation are run there. Three further backbones (Table 3) carry the headline three-way contrast only, which is what a second model can answer: not *how* the effect decomposes—that is a question about the method, and it is settled once—but whether its *sign* survives a change of scale and of pre-training family. All runs train on the same 17k-problem competition-mathematics set. With a training batch of 512 prompts, 110 optimization steps correspond to about 3.2 epochs.

11.2 Validation set, and why it is not the same everywhere

The mathematics-specialized backbones validate on AIME 2024 (30 problems, 32 samples each). AIME 2024 is small: with a per-problem accuracy spread of .1546 measured across our runs, the standard error of its mean is $.1546/\sqrt{30} \approx .0282$. For a backbone that scores near zero on it, every arm lies inside one standard error, so the benchmark cannot separate them.

The sharper problem is not the reported number but the *selection*: we choose each run’s checkpoint by its best validation accuracy, so a benchmark that cannot distinguish arms also cannot distinguish checkpoints, and the selection becomes a draw from noise. The non-mathematics backbones therefore validate and select on MATH500 (500 problems, one sample each), whose standard error is roughly four times smaller.

This buys a measurable comparison at the cost of a protocol that is not identical across backbones. We state it here rather than in a footnote because it constrains what may be read off the results: every contrast we report is *within* a backbone, and no claim compares absolute accuracy between rows of Table 3.

Base or instruction-tuned. The same measurement forces a second protocol difference. The Llama backbone is the instruction-tuned checkpoint; the other three are base models. We began with the base one and scored it under this protocol before training it: it reached 0.020 on MATH500 and 0.024 on grade-school arithmetic. The cause is mechanical rather than mathematical—that checkpoint ships a chat template, which our data pipeline applies, so a model that was never instruction-tuned receives the instruction format and continues the prompt instead of answering it.

The instruction-tuned checkpoint of the same family and the same parameter count, scored under the identical protocol, reaches 0.340 and 0.803 on those two benchmarks. Holding family and scale fixed and changing only the checkpoint moves MATH500 by a factor of seventeen, which is what identifies the failure as one of format rather than of capacity: this backbone can do the arithmetic, and the base checkpoint was not being asked for it in a form it could answer.

What this costs is not a weak row but an absent one. A group of $n = 8$ responses receives zero advantage when its members score alike, so at that solve rate $(1 - 0.020)^8 \approx 0.85$ of groups carry no gradient at all, on a training set harder than the benchmark just quoted. Neither our method nor the baselines can learn from the remainder, and the row would compare two untrained models. We therefore gate a backbone on the fraction of non-degenerate groups on the *training* set, measured against a backbone known to train, rather than on a validation score: a validation set says whether a backbone can be measured, not whether it can learn. On that gate the instruction-tuned checkpoint leaves — of groups informative against — for the 1.5B mathematics backbone, a ratio of —.

Optimization uses GRPO with $n = 8$ responses per prompt, a constant learning rate of $1e-6$ with no warm-up (so that the schedule is independent of the declared total step count), a maximum response length of 3072 tokens, no entropy bonus, and no KL loss. Rollouts are sampled at temperature 1.0 with

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (5)–(15)	nothing — identities	—	—
A1	horizon κ , discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	realized future $\rightarrow$ MTP marginal	mutual information with intervening tokens, Eq. (18)	over-estimate, monotone in k	A3
A3	per-head affine calibration	non-affine residual of A2	least-squares residual	oracle-forecast control arm
A4	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	<i>inherited from the base method</i>
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level entropy-control method we extend.

Backbone	Family	Math-spec.	Validation
1.5B	Qwen2.5-Math	yes	AIME 2024
7B	Qwen2.5-Math	yes	AIME 2024
3B*	Llama 3.2	no	MATH500
7B	Mistral	no	MATH500

Table 3: Backbones. The validation set differs by backbone, deliberately; Section 11.2 says why, and the analysis never compares absolute accuracy across rows. *This row uses the instruction-tuned checkpoint; the other three are base models, and Section 11.2 says why this one cannot be.

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
uniform	.25	tree	support only
permuted	.25	tree	A_H shuffled
STEER-F	.25	tree	A_H as derived

Table 4: The five arms. UNIFORM and PERMUTED run the full pipeline with the forecast’s value destroyed in two different ways, so any gain they show is attributable to the apparatus rather than to the visitation signal.

no nucleus truncation; validation uses temperature 1.0 and top- $p = 0.7$. Every arm uses the same data seed and the same initial checkpoint, which we verify by requiring the step-0 validation scores to agree.

11.3 Method hyperparameters

The forecaster uses $\kappa = 2$ heads with discount $\gamma_H = 0.7$ and the per-head calibration of Eq. (19). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, and the sibling-prefix baseline. The weight map is the base method’s exponential map of Eq. (2), taken un-rescaled, at $\lambda_{\min} = 0.7$, so weights land in $[0.7, 1]$ and can only attenuate. The released configuration of the base method sets $\lambda_{\min} = 0.8$; every other value in that launcher matches it. Ours therefore attenuates the extreme- $|\Omega|$ tokens by up to 30% where theirs stops at 20%, and Table 10 carries an arm at the released value so that the difference is measured rather than as-

sumed.¹ The tree sampler uses a single root, cuts at depths $\{64, 192, 384\}$, and branching factor 2 at each cut, giving $n = 8$ leaves.

11.4 Five arms

The design isolates one question: does the *value* of the forecast matter, or only the apparatus that carries it? A comparison between the base method and the full method changes two things at once (λ and the sampler), so it cannot answer this. We therefore add two arms that keep the entire apparatus — the tree, the forecaster, the support, the weight mapping — and destroy only the information in A_H .

UNIFORM applies the same attenuation to every position in the support, discarding A_H ’s value but keeping its support — this is the implementation of the “raise the mean” hypothesis that Section 9.3 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuf-

¹Our implementation exposes an optional reshaping of $\tilde{\Omega}$ before this map (winsorizing or rank-transforming it); every run reported here uses the identity setting, so the base method’s map sees exactly what it would have seen.

fles the assignment, so the support, the magnitudes, and Eq. (25) all survive while the pairing between a branch and its score is destroyed. Together they bracket the claim: if the visitation signal is doing the work, both controls should behave like the no-forecast arms.

11.5 Metrics

We report acc@32 (mean accuracy over 32 samples) and maj@32 (accuracy of the majority vote over the same 32 samples), and their difference

$$\text{uplift} := \text{maj@32} - \text{acc@32},$$

which measures how much better a set of 32 samples is than its average member — that is, whether correct answers concentrate while incorrect ones scatter. We use uplift rather than pass@32 because the latter does not separate the arms.

Because single validation points are noisy on a 30-problem set, the primary statistic is the mean over the converged window, steps 40–110 (8 validation points per arm). We quote differences in units of the across-problem standard error, $\text{SE} = \text{std@32}/\sqrt{30} \approx 0.027$, which is the correct scale for generalization to new problems. We separately note that the pure re-sampling noise of the protocol — estimated from two validations of identical weights — is about 0.004 in accuracy, an order of magnitude smaller; the two scales answer different questions and we are explicit about which is in use.

One instrumentation caveat propagates into every entropy number we report. The logged actor/entropy is aggregated with a token-mean, so it is $\mathcal{H}$ divided by the mean response length, not $\mathcal{H}$. Since response lengths differ by under 5% across arms, we report the raw quantity and, where the distinction could matter, the length-corrected product as well.

12 Results

All numbers below come from the runs described in Section 11: one benchmark, 110 optimization steps per arm, and the same initial checkpoint, whose step-0 accuracy is identical across arms. Every figure is recomputed from a training log released with this paper. We report our own measurements only; no figure in this section is taken from prior work.

The campaign is not finished. Each arm below rests on 1 seed, and where an arm has more we hold it out of the tables rather than average arms

step	GRPO	STEER	UNIF	PERM	STEER-F
0	.039/.035	.043/.040	.039/.035	.039/.035	.039/.035
10	.075/.127	.064/.109	.070/.124	.062/.126	.078/.148
20	.120/.176	.114/.195	.102/.171	.087/.155	.103/.181
30	.108/.195	.135/.215	.121/.223	.099/.178	.113/.211
40	.144/.236	.115/.187	.119/.201	.116/.197	.141/.230
50	.120/.204	.125/.196	.121/.216	.153/.224	.145/.258
60	.126/.178	.139/.202	.125/.203	.135/.190	.145/.237
70	.130/.200	.128/.200	.129/.197	.132/.227	.147/.257
80	.135/.182	.135/.202	.140/.204	.134/.197	.144/.206
90	.136/.191	.144/.212	.142/.178	.130/.210	.166/.239
100	.132/.171	.146/.207	.134/.192	.144/.212	.157/.222
110	.157/.208	.132/.181	.140/.171	.152/.218	.151/.230

Table 5: AIME24 $\text{acc@32}/\text{maj@32}$ at every validation point. Rows below the rule are the converged window used for all subsequent statistics.

arm	n	acc@32	maj@32	uplift	entropy
GRPO	1	.1350	.1962	.0612	.1491
STEER	1	.1330	.1984	.0654	.1214
UNIFORM	1	.1313	.1953	.0640	.1374
PERMUTED	1	.1370	.2094	.0724	.1190
STEER-F	1	.1495	.2349	.0854	.1210

Table 6: Means over the converged window (steps 40–110, eight validation points per run), averaged over the n seeds *common to all five arms* so that the rows here and the contrasts in Table 7 are drawn from the same runs. *uplift* is $\text{maj@32} - \text{acc@32}$. Entropy is the token-mean quantity discussed in Section 11, i.e. $\mathcal{H}$ divided by mean length.

over different seed sets — a reader subtracting two rows of Table 6 would otherwise get a different number from Table 7. Those extra runs are not wasted: Section 12.8 uses them for the one thing a single seed cannot give, a direct measurement of how far the same configuration moves when it is run again.

12.1 Validation trajectory

Table 5 gives AIME24 accuracy and majority-vote accuracy at every validation point. The four arms that share the tree pipeline start from an identical step-0 score; STEER starts at .043 rather than .039, a gap we attribute to re-sampling noise rather than to a different checkpoint, since two validations of the *same* base weights in our logs differ by .004 in accuracy.

12.2 Converged window

Table 6 averages each arm over steps 40–110.

Two orderings are worth stating before the contrasts. On accuracy, the two arms with no forecast (STEER, UNIFORM) sit *below* plain GRPO, and only the arm carrying A_H is clearly above it. On

contrast	n	Δ_{acc}	t	Δ_{maj}	t
STEER-F – GRPO	1	+.0145	–	+.0386	–
STEER-F – STEER	2	+.0135	+4.50	+.0211	+1.36
STEER-F – UNIF	2	+.0114	+1.65	+.0249	+1.69
STEER-F – PERM	1	+.0125	–	+.0255	–
STEER – GRPO	3	–.0101	–2.33	–.0067	–1.11
UNIF – STEER	3	+.0036	+1.34	–.0031	–3.71

Table 7: Contrasts paired by *seed*: each arm is averaged over its converged window, arms are paired within a seed, and t is the paired t on the n seed-level differences ($\text{df} = n - 1$). Pairing within a seed cancels any per-seed effect common to all five arms. A contrast’s n is the smaller of its two arms’ seed counts.

entropy the ordering is almost reversed: GRPO retains the most entropy and is second-to-last on accuracy, while STEER-F retains among the least and is first.

12.3 Does the forecast’s value matter?

Table 7 averages each arm over its converged window, pairs the arms within a seed, and reports the mean difference together with a paired t on those seed-level differences. The step-by-step version of this statistic, which the single-seed draft reported, is in Appendix D; it answers a question about one trajectory rather than about re-running.

The block structure is what the design was built to expose. Every contrast that changes the *value* of A_H is positive; the one that leaves it alone is not distinguishable from zero. UNIFORM runs the entire apparatus — tree sampler, forecaster, support, weight map — and differs from STEER by +.0036 ($t = +1.34$ over 3 seeds): turning the machinery on and feeding it a constant buys nothing. PERMUTED keeps the machinery *and* the magnitudes of A_H and destroys only the pairing between a branch and its score, and STEER-F still exceeds it by +.0125.

We state the signs, not significance. At the seed counts reached so far — the headline contrast rests on 1 — no accuracy contrast in Table 7 clears a conventional threshold, and the two rows with $n = 1$ carry no t at all. What the table shows is that the sign is the same wherever A_H carries information and absent where it does not; Section 12.8 states what the completed campaign is powered to detect, and the pre-registration fixes the test before the remaining seeds land.

This is the contrast the design was built for. A comparison of STEER-F against STEER alone changes two things at once, and cannot separate the forecast from the sampler that delivers it; the

contrast	Δ_{entropy}	Δ_{acc}
GRPO – STEER	+.0277	+.0020
GRPO – STEER-F	+.0281	–.0145
STEER – STEER-F	+.0004	–.0165

Table 8: Aggregate entropy does not order the arms. The last row is the decisive one: two arms whose converged entropy differs by .0004 differ in accuracy by .0165 and in majority vote by .0365.

two destroyed-forecast arms do separate them.

12.4 Where the gain comes from

The majority-vote gain decomposes into a part explained by per-sample accuracy and a part explained by the sample set’s structure:

$$\underbrace{+.0386}_{\Delta_{\text{maj}}} = \underbrace{+.0145}_{\Delta_{\text{acc}}} + \underbrace{+.0241}_{\Delta_{\text{uplift}}},$$

against GRPO; against STEER the uplift term is +.0076 ($t = +0.61$ over 2 seeds). Against GRPO roughly two thirds of the improvement is not that individual samples are more often right, but that the right ones agree with each other more and the wrong ones agree less. The two backbones of that split are measured on one seed so far, and the split itself is the claim to re-check as the remaining seeds arrive: against STEER the same term is smaller and well inside its own spread.

Where that structural change comes from is a separate question, and the arm that answers it finished after this section was first written. UNIFORM runs the sampler at $\lambda = .25$ with the forecast’s value destroyed and shows no uplift gain, which reads as though the structure belonged to A_H . It does not: the $\lambda = 0$ tree arm — the sampler with the future channel off, STEER’s local damping still in place — reaches an uplift of .0870, against .0654 for STEER and .0854 for STEER-F. On this evidence most of the structural gain is the sampler’s, and Section 12.10 reports it that way. The decomposition above is unaffected — it is measured against GRPO, which has no tree — but the attribution is not what the five-arm design alone could see.

12.5 Entropy is not the channel

Table 8 makes three separate points. First, both intervened arms hold *less* entropy than plain GRPO, so whatever they are doing, it is not suppressing the collapse. Second, the arm with the most entropy is not the arm with the most accuracy. Third, and decisively, STEER and STEER-F converge to the

same entropy to within .0004 — the same number of paths remain open — and yet differ by .0165 in accuracy. The quantity that separates them is not how much entropy survives but which branches it survives on.

This is what Section 9.3 predicted. A_H sums to zero over a sibling set, so it cannot move the mean damping; measured over the whole run the mean token weight is 0.999 in every arm. An intervention that is mean-preserving by construction *should* leave the entropy trajectory alone, and it does.

12.6 Mechanism diagnostics

Section 9 bounds the fraction of positions that can carry a non-zero A_H at $2(n-1)/T$ for any sampler. With $n = 8$ and a mean response length of 984 tokens the bound is .0142. Measured over training, the three tree arms sit at .0136 (UNIFORM), .0129 (PERMUTED) and .0120 (STEER-F), i.e. at 84–96% of a ceiling that no sampler can exceed. A plain-rollout run with the same λ reaches .003. As Section 8 notes, this count admits float32 round-off and is an upper bound; the runs reported here predate the round-off-free counter, so we quote the loose figure and label it as such.

Two readings follow. The tree sampler is close to saturating what the combinatorics allow, so there is little headroom in scheduling more cuts; and the intervention is extremely sparse — about one position in eighty — which makes the size of the effect in Table 7 the surprising part rather than the small support.

12.7 Is the future term a local term?

Section 6 reports a calibration that suppresses every head past the first, leaving $\hat{H}_{\text{togo}}$ dominated by one step ahead. That invites the sharpest objection to the paper: if the forecast reaches one step, is A_H anything a purely local quantity could not have produced? The question is answerable without training anything. We apply the *same* operator — the same sibling baseline, mask and grouping — to two inputs: the calibrated forecast, and the policy’s own next-token entropy read at the same offset, the state after the branch token. Any difference between the two is the forecaster’s contribution and nothing else. We report it on the sibling support, where A_H can be non-zero at all; outside it both are identically zero and a statistic over all positions measures the support rather than the forecast.

The answer is that the forecaster is not earning its cost. Over 464 branch points from 4 val-

idation prompts, the two advantages correlate at $r = +0.917$ (Spearman $+0.937$), the local twin explains $R^2 = 0.840$ of the treatment, and the two agree in sign 0.921 of the time — and the sign is what the token weights are monotone in. The levels correlate less than the advantages ($+0.881$), which is the wrong way round for an artefact of scale.

The mechanism is one the paper had already measured without connecting it. With $\kappa = 2$ and $\gamma_H = 0.7$ the calibration instantiates to $\hat{H}_{\text{togo}} = 0.72 H_1 + 0.06 H_2 + 0.11$, the sibling baseline removes the constant, and the first head is built to reproduce the policy’s own next-token distribution — its final layer is zero-initialised and its fitted $a_1 = 1.03$ says it stayed there. So A_H should be close to $\gamma_H a_1 = 0.72$ times the local twin, and the measured magnitudes are 0.3445 against 0.4647, a ratio of 0.741. A_H is very nearly a scalar multiple of a quantity the trainer already computes, and Eq. (23) divides the scale out through $\rho = \text{rms}\{\text{visit}\}$.

Two things this does *not* say. It does not touch Eq. (8), which is an identity, nor Section 4.4: the local twin is not Ω . Ω carries no sibling baseline and no tree, vanishes at a uniform branch point, and the twin exists only because the tree supplies siblings to difference against. What the measurement indicts is the *forecasting apparatus* — the MTP heads, their warm-up and their calibration — not the visitation channel. Read alongside Section 12.10, where the sampler alone recovers .1439 against STEER’s .1330 while the forecast adds the remainder, two independent measurements now point the same way.

The caveats are the ones a single diagnostic run carries: 4 prompts at one validation step of the base policy, on independently sampled rollouts, whose sibling support therefore sits early in the response rather than at the scheduled branch depths. The head-free arm of Table 10 is the direct test, and it is cheaper than the arm it would replace: ORACLE reads H_{togo} from the policy’s realised entropy, which verl computes in the same pass, so it carries none of the overhead in Section 12.13.

12.8 Seeds, not steps

Every statistic above treats the seed as the unit of analysis: each run is first collapsed to one number by averaging its converged window, the arms are then paired within a seed, and the test is a one-sample t on the paired differences with $\text{df} = n - 1$. Pairing within a seed matters because a seed carries

effects common to all five arms — the data order, and on a split campaign the machine a run landed on — and those cancel exactly in a difference. A contrast’s n is the smaller of its two arms’ seed counts, which is why the table reports n per row rather than once.

The alternative statistic, a paired t over the eight window steps *within* one run, measures whether a gap is stable along a single trajectory. That is a different question from whether it survives re-running, and we report it only in Appendix D so the two are not confused.

Run-to-run spread is itself worth reporting, and with one seed per arm it is the number that decides how much the rest of this section is worth. Two arms have now finished twice. GRPO’s converged-window accuracy moves from .1350 to .1560 across its 3 seeds — a spread of .0210, against a headline STEER-F – GRPO gap of +.0145; on majority vote the spread is .0346 against +.0386. PERMUTED moves from .1370 to .1384 across its 2, a spread of .0014, roughly six times smaller.

That disagreement is itself the result. A range between two draws is not a standard error, and two such ranges differing by a factor of six do not bound a third arm’s. We therefore read the larger one as the operative figure and decline the smaller, which means no contrast in this paper should be compared against a noise floor quoted to one significant figure. On that conservative reading the gap is larger than the spread on both metrics, and on majority vote by more than a factor of two. But it is the *same order* as the spread, which is precisely the regime in which a single seed cannot decide the question: the comparison we are making is between one draw from each arm, and the arm we can observe twice moved by more than half the effect we are claiming. For scale at the other end, two validations of the *same* weights on these 30 questions differ by about .004, so seed variation is the larger term by roughly an order of magnitude and any comparison quoted at single-run precision is quoting noise. We therefore state the ordering and the mechanism as what this experiment establishes, and the effect size as what the remaining seeds are for.

12.9 Six benchmarks

A single 30-question benchmark cannot separate a method from a lucky draw, so each run’s selected checkpoint is re-evaluated on six mathematics benchmarks (Table 9) under the protocol

arm	AIME24	AIME25	AMC23	MATH500	Minerva	Olympiad	mean
							–

Table 9: Six-benchmark evaluation of each arm’s selected checkpoint, averaged over seeds. The rightmost column is the unweighted mean, the headline figure of the post-hoc evaluation.

arm	what changes	acc@32	maj@32
<i>the damping coefficient</i>			
$\lambda = 0.1$	weaker	.1365	.2360
$\lambda = 0.25$	as used	.1495	.2349
$\lambda = 0.5$	stronger	.1444	.2265
<i>the sampler, with the forecast switched off</i>			
$\lambda = 0$, tree	tree only	.1439	.2309
STEER	neither	.1330	.1984
<i>the forecaster, replaced by what the trainer already has</i>			
ORACLE	no heads	–	–
<i>the base method at its released damping strength</i>			
STEER, $\lambda_{\min}=0.8$	weaker	–	–
<i>relaxed ratio clipping ($\epsilon_{hi}=5$, $\epsilon_{lo}=0.99$)</i>			
STEER-F	—	.1516	.2304
STEER	—	–	–
<i>other advantage estimators</i>			
RLOO, STEER-F	—	–	–
RLOO, STEER	—	–	–
OPO, STEER-F	—	–	–
OPO, STEER	—	–	–

Table 10: Follow-up ablations, seed 1 only, converged window. Each block isolates one element of the design.

of Section 11: AIME24, AIME25 and AMC23 at avg@32, MATH500, Minerva and OlympiadBench at avg@1.

The quantity fixed in advance as the headline here is not any single cell but *direction consistency*: in how many of the six benchmarks the ordering observed on AIME24 is preserved. Four or better of six we declared support, three equivocal, two or fewer evidence against. The observed count is – of six for STEER-F > UNIFORM, and – of six for STEER-F > PERMUTED. This is the test that distinguishes a method from an artifact of one question set, and it is reported whatever it returns.

12.10 How much of the design is load-bearing

Table 10 varies one element at a time, at a single seed. Every comparison in it is therefore $n = 1$: these rows describe a mechanism, and none of them carries an error bar or licenses an effect size.

The coefficient. λ scales the visitation term before the two channels are combined in Eq. (23). The value used throughout was not tuned on these runs, and the sweep exists to say whether the result rests on a knife edge. A flat response across 0.1–0.5 would say the method is insensitive to it;

a peak at 0.25 with both neighbours below would say it is a tuned quantity and must be reported as one.

The damping strength. We measure STEER below GRPO by $-.0101$ over 3 seeds, which the base method’s own results contradict. The one configuration difference we have found is $\lambda_{\min}$: 0.7 here against 0.8 in the released script, so our STEER arm applies one and a half times their maximum attenuation. This arm runs $\lambda = 0$ — stock STEER — at their value, and separates a failure to reproduce from an over-damped control. Note what it cannot settle on its own: the STEER-F arms also run at 0.7, so a difference here would mean the $\lambda_{\min}$ shared by both sides of every contrast is not neutral, and the comparison would have to be re-based rather than corrected.

The sampler. This is the cell no other arm supplies. Tree rollouts make a group share prefixes, which changes the GRPO advantage estimate itself, quite apart from any damping. UNIFORM and PERMUTED both run at $\lambda = .25$ and so cannot separate the sampler from the correction. The $\lambda = 0$ tree arm switches the forecast off while leaving the sampler in place, and is the direct answer to “is tree rollout simply a better sampler”. Read against STEER, which has neither, its difference is the sampler’s own contribution; read against STEER-F, what remains is the forecast’s.

The answer is not the one the rest of this paper was written expecting. The sampler alone reaches .1439 against STEER’s .1330, and .2309 against .1984 — about two thirds of STEER-F’s accuracy margin over STEER and the large majority of its majority-vote margin. Its uplift, .0870, is not below STEER-F’s .0854. What remains for the forecast is the accuracy gap between .1439 and .1495.

Two things keep this from being decisive, and neither rescues the original reading. Every cell here is one run, and the spread across this block is about twice the run-to-run spread measured in Section 12.8 — suggestive, not settled. And the λ sweep is not monotone: $\lambda = .1$ sits below $\lambda = 0$ on accuracy, which is what noise looks like at this sample size. We report the block as measured and do not reconcile it. The five-seed campaign of Appendix C decides whether the sampler or the forecast carries the structural term; until then, UNIFORM and PERMUTED establish that the forecast’s *value* matters at fixed λ , and nothing in this paper establishes that the tree does not.

arm	steps	wall clock	acc@32
STEER-F	110	—	.1495
GRPO, step-matched	110	—	.1350
GRPO, compute-matched	—	—	—

Table 11: Equal steps versus equal compute. The matched step is read from the integrated per-step time of the long GRPO run, on the same hardware as its STEER-F reference; times measured on a different machine are not comparable and are not mixed in.

Clipping and the advantage estimator. The relaxed-clipping setting is where entropy interventions are supposed to matter most, because ratio clipping otherwise removes much of the update a token weight would have scaled. The RLOO and OPO rows test whether the construction depends on GRPO specifically: the sibling-prefix baseline of Eq. (15) depends only on the group sharing a prefix, which those estimators also provide, so the method should transfer unchanged. Each is reported beside its own base-method control, so every comparison stays within one estimator.

12.11 Compute-matched comparison

STEER-F costs more per step than GRPO, almost entirely in the MTP forward pass, so an equal-step comparison hands it more wall clock. The honest comparison holds compute fixed instead. We run GRPO to — steps, integrate its measured per-step time, and read its accuracy at the last step whose cumulative time fits inside what STEER-F spent reaching step 110 — rather than assuming a cost ratio.

Both readings are reported. If the gain survives the compute-matched comparison it is a gain; if it does not, the method buys accuracy with time and should be described that way.

12.12 Does the sign survive a change of backbone?

Everything above is one model. A reviewer’s first question is whether the effect is a property of the method or of that model, and the honest form of the answer is narrow: on three further backbones we run only GRPO, the base method, and ours, and we ask whether the *ordering* holds. Absolute accuracies are not comparable across rows—the backbones differ in scale, in pre-training family, in whether mathematics was specialized for, and (Section 11.2) in which benchmark selects the checkpoint—so the table reports each backbone’s

	GRPO	base	ours	Δ GRPO	Δ base
1.5B Qwen-Math	.1350	.1330	.1495	+.0145	+.0135
7B Qwen-Math	-	-	-	-	-
3B Llama	-	-	-	-	-
7B Mistral	-	-	-	-	-

Table 12: The headline contrast on four backbones, seed 1, converged window. The two rightmost columns are within-backbone differences; columns are not averaged, because the rows are not on a common scale. The Llama row uses the instruction-tuned checkpoint (Section 11.2); the other three are base models.

own three numbers and the within-backbone differences, and nothing is averaged down a column.

Two questions are separable here, and the design answers them one at a time. The 7B model shares its tokenizer, architecture and pre-training corpus with the 1.5B, so a difference between those two rows is about *scale*. The Llama and Mistral backbones share none of that, so a difference there is about *family*, and in particular about whether the effect needs a base policy that is already sharp on mathematics. Our account predicts that the visitation channel matters wherever the rollout distribution has branch points that differ in what they lead to, which is a property of the task rather than of the pre-training corpus; a backbone on which the sign reverses would be evidence against that account, not merely a weaker result.

The statistic we report is the sign of Δ_{GRPO} , and how many of the four backbones share it. With one seed per cell, a single backbone’s difference carries about the same uncertainty as a single seed on the primary backbone (Section 12.8), so no individual row is decisive; the claim rests on agreement across rows. We report the count, $-$ of 4, and treat a split result as a limit on generality rather than as noise to be averaged away.

12.13 Cost

Within the runs executed on identical hardware, STEER takes 833s per optimization step and STEER-F takes 1516s, an overhead of 82%. The UNIFORM and PERMUTED arms cost 1480 and 1492s, confirming that the cost is the tree rollout plus the MTP forward pass and not the A_H computation, which is a subtraction over a mask. We do not report a GRPO timing because that arm ran on a different machine.

12.14 A note on how these numbers were selected

We report window means rather than the best checkpoint, and the difference is not cosmetic. Selecting the maximum over the twelve validation points inflates every arm by a similar amount: $+.013$ for STEER, $+.011$ for UNIFORM, $+.016$ for PERMUTED and $+.017$ for STEER-F, relative to the window mean. Since a single AIME24 measurement carries a standard error of $.0282$ across problems, a maximum over a dozen noisy points is a biased statistic whose bias grows with the number of checkpoints compared. The arm ordering happens to be the same under both rules here, but we regard the window mean as the defensible one and use it throughout.

13 Conclusion

The gradient of trajectory entropy has two terms, and the token-level entropy controls now in use estimate one of them. We showed that the omission is structural: the estimated term vanishes exactly where the omitted term is largest, and it vanishes for a correct reason, so the gap cannot be closed by improving the estimator. We then showed that the omitted term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy, that a parallel forecaster estimates it for the price of one forward pass with an error whose sign and monotonicity are known in closed form, and that a tree-structured rollout supplies the baseline it requires. The resulting correction is mean-preserving by construction: it does not slow entropy collapse, it redirects it.

Limitations

A single validation benchmark, and what our t -statistics do not cover. Every claim rests on AIME 2024, which has 30 problems. A single measurement of one arm carries an across-problem standard error of $.0282$, so the gaps in Table 7 are a fraction of one problem in absolute terms. The paired t we report is computed over the eight validation steps of one run: it establishes that a gap is stable across the converged window, and nothing more. The statistic that would speak to generalization — the across-problem standard error of the *paired* difference, in which per-problem difficulty cancels between two arms evaluated on the same 30 questions — is not recoverable from our logs, which record per-benchmark means and standard

deviations rather than per-problem scores. We state this rather than substitute a number we cannot compute. A multi-benchmark evaluation, and logging per-problem scores so the paired error can be estimated, are the two things that would settle it.

A single seed, and a gap the same order as the noise on it. Every arm rests on one data seed, so no contrast in this paper carries a seed-level test. This is the limitation that bounds all the others. We can now put a number on it rather than an argument, though not yet a stable one: of the two arms run twice, GRPO moved .0210 in converged-window accuracy between its seeds and PERMUTED moved .0014, a sixfold disagreement between two estimates of the same quantity. Against the larger, and a headline gap of +.0145, the gap is the bigger of the two, and on majority vote by more than a factor of two, but not by the margin that would let a single draw from each arm settle it — and the sixfold spread between the two estimates means the floor itself is not yet measured. The across-problem standard error we quote elsewhere is a different quantity and does not bound this one. Rollout sampling is not seeded across processes, so a re-run of the same configuration is a second draw rather than a restoration — which is what makes the repeat measurable, and what makes it necessary. The pre-registered five-seed campaign in Appendix C is designed to settle the effect size; what this paper establishes without it is the ordering of the arms and the mechanism that produces it, since the three controls sit at zero while only the arm carrying A_H moves.

The forecaster does not earn its cost. The calibration in Eq. (19) suppresses heads beyond the first, so $\hat{H}_{\text{togo}}$ is dominated by a single step ahead. This is the predicted behavior of Eq. (18) rather than a bug, but Section 12.7 measures the consequence and it is worse than a short horizon: put through the same sibling baseline, the policy’s own next-token entropy reproduces A_H at $r = +0.917$ and $R^2 = 0.840$, with the magnitudes in the ratio 0.741 that γ_{Ha_1} predicts. On that evidence the MTP heads, their warm-up and their calibration are machinery around a quantity the trainer already computes, and they carry the overhead of Section 12.13 to do it. We report the ORACLE arm of Table 10 as the direct test. What the measurement does not reach is the visitation channel itself: the local twin is not Ω , and it exists only because the tree supplies siblings to difference against. A

forecaster whose error did not grow with k would test the idea more sharply; a sequential forecaster is the obvious comparison and we have not run it.

The support is small by construction. Section 9 bounds the fraction of positions carrying a non-zero A_H at $2(n - 1)/T$, on the order of a percent for our settings. Whether the effect scales with n , or with a branching schedule chosen adaptively rather than at fixed depths, is untested.

An assumption with no remedy. Approximation B2 — that a token’s update does not change the conditional distribution at other positions — is false in a network with shared parameters. It is inherited from the base method rather than introduced here, but our construction adds a second estimator that relies on it.

One prediction of the analysis is unmeasured. Section 4.4 argues that the two channels are anti-correlated. The prefactor half of that claim is analytic and the A_H half is inferred from a correlation with branch diversity; we have not measured their *product* at real sibling-divergence positions in a live rollout. Until that measurement exists, Table 1 is a prediction rather than an observation.

Scope of the backbone evidence. The full decomposition—the five arms, the ablations, the mechanism diagnostics—rests on one 1.5B model. The three further backbones (Section 12.12) carry the three-way contrast at one seed each, which establishes the sign and not its size, and they stop at 7B: we have no evidence about models an order of magnitude larger, where the base policy is sharper and the relative size of the two channels may differ. Family and scale are also not fully crossed—the mathematics backbones span 1.5B to 7B and the others sit at 3B and 7B—so what the latter test is whether the sign survives a different pre-training corpus, not whether it survives one at 8B or beyond. Those rows further differ from the mathematics ones in more than family. They are selected on a different benchmark (Section 11.2), and the Llama row is instruction-tuned where the other three are base models, which was forced rather than chosen: the base checkpoint left almost every training group degenerate. Each difference is what makes the row measurable at all, and each is a reason no claim here compares accuracy across backbones—only the sign of a within-backbone contrast is read.

Ethics Statement

This work studies an optimization-time regularizer for reinforcement learning on mathematics problems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation set is a public examination.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are not specific to mathematics, and the same mechanism would apply to a reward model encoding any objective, including a harmful one; we note that our method inherits, and does not mitigate, whatever the reward function encodes. The computational cost of the method (a forward pass through a set of small prediction heads at every optimization step) is a real environmental cost, which we quantify in Section 12.13 so that it can be weighed against the benefit.

References

Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, , and Sara Hooker. 2024. Back to basics: Revisiting reinforce-style optimization for learning from human feedback in llms. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12248–12267.

Shengnan An, Xunliang Cai, Xuezhi Cao, Xiaoyu Li, Yehao Lin, Junlin Liu, Xinxuan Lv, Dan Ma, Xuanlin Wang, Ziwen Wang, , et al. 2025. Amo-bench: Large language models still struggle in high school math competitions. *ArXiv preprint arXiv:2510.26768*.

Anonymous. 2025. Rethinking entropy regularization in large reasoning models. *ArXiv preprint arXiv:2509.25133*.

Anthropic. 2025. Introducing claude opus 4.5. <https://www.anthropic.com/news/claude-opus-4-5>.

Sikai Bai, Haoxi Li, Jie Zhang, Yongjiang Liu, , and Song Guo. 2026. Ttvs: Boosting self-exploring reinforcement learning via test-time variational synthesis. *ArXiv preprint arXiv:2604.08468*.

Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwei Peng, Jason D. Lee, Deming Chen, and Tri Dao. 2024. Medusa: Simple LLM inference acceleration framework with multiple decoding heads. *ArXiv preprint arXiv:2401.10774*.

Yupeng Chang, Yi Chang, , and Yuan Wu. 2026. Balora: Bias-alleviating low-rank adaptation to mitigate catastrophic inheritance in large language models. In *The Fourteenth International Conference on Learning Representations*.

Yupeng Chang, Chenlu Guo, Yi Chang, , and Yuan Wu. 2025. Lora-mgpo: Mitigating double descent in low-rank adaptation via momentum-guided perturbation optimization. in *findings of the association for computational linguistics: Emnlp 2025*, pages 648–659.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, , et al. 2024. A survey on evaluation of large language models. *acm transactions on intelligent systems and technology*, 15(3):1–45.

Jiawei Chen, Hande Dong, Xiang Wang, Fuli Feng, Meng Wang, , and Xiangnan He. 2023. Bias and debias in recommender system: A survey and future directions. *acm transactions on information systems*, 41(3):1–39.

Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Wayne Xin Zhao, Zhenliang Zhang, , and Furu Wei. 2025. Reasoning with exploration: An entropy perspective. *ArXiv preprint arXiv:2506.14758*.

Xiangxiang Chu, Hailang Huang, Xiao Zhang, Fei Wei, , and Yong Wang. 2025. Gpg: A simple and strong reinforcement learning baseline for model reasoning. *ArXiv preprint arXiv:2504.02546*.

Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, , et al. 2025a. Process reinforcement through implicit rewards. *ArXiv preprint arXiv:2502.01456*.

Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, , et al. 2025b. The entropy mechanism of reinforcement learning for reasoning language models. *ArXiv preprint arXiv:2505.22617*.

Yu Cui, Feng Liu, Jiawei Chen, Canghong Jin, Xingyu Lou, Changwang Zhang, Jun Wang, Yuegang Sun, , and Can Wang. 2025c. Hatllm: Hierarchical attention masking for enhanced collaborative modeling in llm-based recommendation. *ArXiv preprint arXiv:2510.10955*.

Jia Deng, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, , and Ji-Rong Wen. 2025. Decomposing the entropy-performance exchange: The missing keys to unlocking effective reinforcement learning. *ArXiv preprint arXiv:2508.02260*.

Chongming Gao, Kexin Huang, Jiawei Chen, Yuan Zhang, Biao Li, Peng Jiang, Shiqi Wang, Zhong Zhang, , and Xiangnan He. 2023a. Alleviating matthew effect of offline reinforcement learning in interactive recommendation. In *Proceedings of the*

1525	46th international ACM SIGIR conference on re-	Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul	1577
1526	search and development in information retrieval,	Arora, Steven Basart, Eric Tang, Dawn Song, , and Ja-	1578
1527	pages 238–248.	cob Steinhardt. 2021. Measuring mathematical prob-	1579
1528	Chongming Gao, Shiqi Wang, Shijun Li, Jiawei Chen,	lem solving with the math dataset. ArXiv preprint	1580
1529	Xiangnan He, Wenqiang Lei, Biao Li, Yuan Zhang, ,	arXiv:2103.03874.	1581
1530	and Peng Jiang. 2023b. Cirs: Bursting filter bubbles	Jian Hu, Jason Klein Liu, Haotian Xu, , and Wei Shen.	1582
1531	by counterfactual interactive recommender system.	2025a. Reinforce++: An efficient rlhf algorithm with	1583
1532	acm transactions on information systems, 42(1):1–	robustness to both prompt and reward models. ArXiv	1584
1533	27.	preprint arXiv:2501.03262, 1(3):5.	1585
1534	Google. 2025. A new era of intelligence with gemini 3.	Jingcheng Hu, Yinmin Zhang, Qi Han, Daxin Jiang,	1586
1535	https://blog.google/products/gemini/gemini-3/ .	Xiangyu Zhang, , and Heung-Yeung Shum. 2025b.	1587
1536	Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri,	Open-reasoner-zero: An open source approach to	1588
1537	Abhinav Pandey, Abhishek Kadian, Ahmad Al-	scaling up reinforcement learning on the base model.	1589
1538	Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten,	ArXiv preprint arXiv:2503.24290.	1590
1539	Alex Vaughan, , et al. 2024. The llama 3 herd of	Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Day-	1591
1540	models. ArXiv preprint arXiv:2407.21783.	iheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang,	1592
1541	Aaron Grattafiori et al. 2024. The llama 3 herd of mod-	Bowen Yu, Keming Lu, , et al. 2024. Qwen2.5-coder	1593
1542	els. ArXiv preprint arXiv:2407.21783.	technical report. ArXiv preprint arXiv:2409.12186.	1594
1543	Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song,	Zed Industries. 2025. zeta. https://huggingface.co/datasets/zed-industries/zeta .	1595
1544	Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma,		1596
1545	Peiyi Wang, Xiao Bi, , et al. 2025. Deepseek-r1: In-	Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richard-	1597
1546	centivizing reasoning capability in llms via reinforce-	son, Ahmed El-Kishky, Aiden Low, Alec Helyar,	1598
1547	ment learning. ArXiv preprint arXiv:2501.12948.	Aleksander Madry, Alex Beutel, Alex Carney, , et al.	1599
1548	Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, , and	2024. Openai o1 system card. ArXiv preprint	1600
1549	Sergey Levine. 2018. Soft actor-critic: Off-policy	arXiv:2412.16720.	1601
1550	maximum entropy deep reinforcement learning with	Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia	1602
1551	a stochastic actor. in international conference on ma-	Yan, Tianjun Zhang, Sida Wang, Armando Solar-	1603
1552	chine learning, pages 1861–1870. pmlr.	Lezama, Koushik Sen, , and Ion Stoica. 2024. Live-	1604
1553	Yaru Hao, Li Dong, Xun Wu, Shaohan Huang, Zewen	codebench: Holistic and contamination free eval-	1605
1554	Chi, , and Furu Wei. 2025. On-policy rl with optimal	uation of large language models for code. ArXiv	1606
1555	reward baseline. ArXiv preprint arXiv:2505.23585.	preprint arXiv:2403.07974.	1607
1556	Zhezhen Hao, Hong Wang, Haoyang Liu, Jian Luo,	Albert Q. Jiang et al. 2023. Mistral 7b. ArXiv preprint	1608
1557	Jiarui Yu, Hande Dong, Qiang Lin, Can Wang, and	arXiv:2310.06825.	1609
1558	Jiawei Chen. 2026. Rethinking entropy interven-	Yihang Jiang, Xiaoyang Li, Guangxu Zhu, Hang Li,	1610
1559	tions in RLVR: An entropy change perspective. In	Jing Deng, Kaifeng Han, Chao Shen, Qingjiang Shi,	1611
1560	<i>Proceedings of the 64th Annual Meeting of the As-</i>	, and Rui Zhang. 2023. 6g non-terrestrial networks	1612
1561	<i>sociation for Computational Linguistics (Volume 1:</i>	enabled low-altitude economy: Opportunities and	1613
1562	<i>Long Papers)</i> , pages 31105–31133.	challenges. ArXiv preprint arXiv:2311.09047.	1614
1563	Andre He, Daniel Fried, , and Sean Welleck. 2025a. Re-	Nathan Lambert, Jacob Morrison, Valentina Pyatkin,	1615
1564	warding the unlikely: Lifting grpo beyond distribu-	Shengyi Huang, Hamish Ivison, Faeze Brahman,	1616
1565	tion sharpening. ArXiv preprint arXiv:2506.02355.	Lester James V Miranda, Alisa Liu, Nouha Dziri,	1617
1566	Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu,	Shane Lyu, , et al. 2024. Tulu 3: Pushing frontiers in	1618
1567	Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yu-	open language model post-training. ArXiv preprint	1619
1568	jie Huang, Yuxiang Zhang, , et al. 2024. Olympiad-	arXiv:2411.15124.	1620
1569	bench: A challenging benchmark for promoting agi	Aitor Lewkowycz, Anders Andreassen, David Do-	1621
1570	with olympiad-level bilingual multimodal scientific	han, Ethan Dyer, Henryk Michalewski, Vinay Ra-	1622
1571	problems. ArXiv preprint arXiv:2402.14008.	masesh, Ambrose Slone, Cem Anil, Imanol Schlag,	1623
1572	Jujie He, Jiakai Liu, Chris Yuhao Liu, Rui Yan, Chaojie	Theo Gutman-Solo, , et al. 2022. Solving quan-	1624
1573	Wang, Peng Cheng, Xiaoyu Zhang, Fuxiang Zhang,	titative reasoning problems with language models.	1625
1574	Jiacheng Xu, Wei Shen, , et al. 2025b. Skywork	advances in neural information processing systems,	1626
1575	open reasoner 1 technical report. ArXiv preprint	35:3843–3857.	1627
1576	arXiv:2505.22312.	Jia Li, Edward Beeching, Lewis Tunstall, Ben Lip-	1628
		kin, Roman Soletskyi, Shengyi Huang, Kashif Rasul,	1629
		Longhui Yu, Albert Q Jiang, Ziju Shen, , et al. 2024.	1630
		Numinamath: The largest public dataset in ai4maths	1631

1632	with 860k pairs of competition math problems and	John Schulman, Filip Wolski, Prafulla Dhariwal, Alec	1685
1633	solutions. hugging face repository, 13(9):9.	Radford, , and Oleg Klimov. 2017. Proximal	1686
1634	Mengdi Li, Jiaye Lin, Xufeng Zhao, Wenhao Lu,	policy optimization algorithms. ArXiv preprint	1687
1635	Peilin Zhao, Stefan Wermt, , and Di Wang. 2025a.	arXiv:1707.06347.	1688
1636	Curriculum-rlaif: Curriculum alignment with rein-	Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu,	1689
1637	forcement learning from ai feedback. ArXiv preprint	Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan	1690
1638	arXiv:2505.20075.	Zhang, YK Li, Yang Wu, , et al. 2024. Deepseek-	1691
1639	Qingbin Li, Rongkun Xue, Jie Wang, Ming Zhou, Zhi	math: Pushing the limits of mathematical reason-	1692
1640	Li, Xiaofeng Ji, Yongqi Wang, Miao Liu, Zheming	ing in open language models. ArXiv preprint	1693
1641	Yang, Minghui Qiu, , et al. 2025b. Cure: Critical-	arXiv:2402.03300.	1694
1642	token-guided re-concatenation for entropy-collapse	Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin	1695
1643	prevention. ArXiv preprint arXiv:2508.11016.	Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin	1696
1644	Siyi Lin, Chongming Gao, Jiawei Chen, Sheng Zhou,	Lin, , and Chuan Wu. 2025. Hybridflow: A flexi-	1697
1645	Binbin Hu, Yan Feng, Chun Chen, , and Can Wang.	ble and efficient rlhf framework. In Proceedings of	1698
1646	2025. How do recommendation models amplify pop-	the Twentieth European Conference on Computer	1699
1647	ularity bias? an analysis from the spectral perspective.	Systems, pages 1279–1297.	1700
1648	In Proceedings of the Eighteenth ACM International	Yuda Song, Julia Kempe, , and Remi Munos. 2025.	1701
1649	Conference on Web Search and Data Mining, pages	Outcome-based exploration for llm reasoning. ArXiv	1702
1650	659–668.	preprint arXiv:2509.06941.	1703
1651	Aixin Liu, Aoxue Mei, Bangcai Lin, Bing Xue, Bingx-	Hongze Tan and Jianfei Pan. 2025. Gtpo and grpo-s:	1704
1652	uan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang,	Token and sequence-level reward shaping with policy	1705
1653	Chaofan Lin, Chen Dong, , et al. 2025a. Deepseek-	entropy. ArXiv preprint arXiv:2508.04349.	1706
1654	v3. 2: Pushing the frontier of open large language	Haozhe Wang, Qixin Xu, Che Liu, Junhong Wu,	1707
1655	models. ArXiv preprint arXiv:2512.02556.	Fangzhen Lin, , and Wenhua Chen. 2025a. Emergent	1708
1656	Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin	hierarchical reasoning in llms through reinforcement	1709
1657	Dong, Yejin Choi, Jan Kautz, , and Yi Dong. 2025b.	learning. ArXiv preprint arXiv:2509.03646.	1710
1658	Prorl: Prolonged reinforcement learning expands rea-	Hong Wang, Zhezheng Hao, Jian Luo, Chenxing Wei,	1711
1659	soning boundaries in large language models. ArXiv	Yao Shu, Lei Liu, Qiang Lin, Hande Dong, , and	1712
1660	preprint arXiv:2505.24864.	Jiawei Chen. 2025b. Scheduling your llm reinforce-	1713
1661	Ziyu Liu, Yuhang Zang, Shengyuan Ding, Yuhang	ment learning with reasoning trees. ArXiv preprint	1714
1662	Cao, Xiaoyi Dong, Haodong Duan, Dahua Lin, ,	arXiv:2510.24832.	1715
1663	and Jiaqi Wang. 2025c. Spark: Synergistic policy	Jiakang Wang, Runze Liu, Fuzheng Zhang, Xiu Li, ,	1716
1664	and reward co-evolving framework. ArXiv preprint	and Guorui Zhou. 2025c. Stabilizing knowledge,	1717
1665	arXiv:2509.22624.	promoting reasoning: Dual-token constraints for rlvr.	1718
1666	Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi,	ArXiv preprint arXiv:2507.15778.	1719
1667	William Y Tang, Manan Roongta, Colin Cai, Jeffrey	Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shix-	1720
1668	Luo, Tianjun Zhang, Li Erran Li, , et al. 2025. Deep-	uan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin	1721
1669	scaler: Surpassing o1-preview with a 1.5 b model by	Yang, Zhenru Zhang, , et al. 2025d. Beyond the	1722
1670	scaling rl. notion blog.	80/20 rule: High-entropy minority tokens drive effec-	1723
1671	Shichao Ma, Zhiyuan Ma, Ming Yang, Xiaofan Li, Xing	tive reinforcement learning for llm reasoning. ArXiv	1724
1672	Wu, Jintao Du, Yu Cheng, Weiqiang Wang, Qiliang	preprint arXiv:2506.01939.	1725
1673	Liu, Zhengyang Zhou, , et al. 2026. Tspo: Break-	Ziqing Wang, Yibo Wen, William Pattie, Xiao Luo,	1726
1674	ing the double homogenization dilemma in multi-	Weimin Wu, Jerry Yao-Chieh Hu, Abhishek Pandey,	1727
1675	turn search policy optimization. ArXiv preprint	Han Liu, , and Kaize Ding. 2025e. Polo: Pref-	1728
1676	arXiv:2601.22776.	erenceguided multi-turn reinforcement learning for	1729
1677	Volodymyr Mnih, Adria Puigdomenech Badia, Mehdi	lead optimization. ArXiv preprint arXiv:2509.21737.	1730
1678	Mirza, Alex Graves, Timothy Lillicrap, Tim Harley,	Fang Wu, Weihao Xuan, Ximing Lu, Zaid Harchaoui, ,	1731
1679	David Silver, , and Koray Kavukcuoglu. 2016. Asyn-	and Yejin Choi. 2025a. The invisible leash: Why	1732
1680	chronous methods for deep reinforcement learning. in	rlvr may not escape its origin. ArXiv preprint	1733
1681	international conference on machine learning, pages	arXiv:2507.14843.	1734
1682	1928–1937. pmlr.	Fei Wu, Zhenrong Zhang, Qikai Chang, Jianshu Zhang,	1735
1683	OpenAI. 2025. Introducing gpt-5.2. https://openai.com/index/introducing-gpt-5-2 .	Quan Liu, , and Jun Du. 2026. Step potential advan-	1736
1684		tage estimation: Harnessing intermediate confidence	1737
		and correctness for efficient mathematical reasoning.	1738
		ArXiv preprint arXiv:2601.03823.	1739

1740	Junkang Wu, Kexin Huang, Jiancan Wu, An Zhang,	reinforcement learning for large reasoning models.	1796
1741	Xiang Wang, , and Xiangnan He. 2025b. Quantile advantage estimation for entropy-safe reasoning. ArXiv preprint arXiv:2509.22611.	ArXiv preprint arXiv:2509.08827.	1797
1742			
1743			
1744	Zhenhe Wu, Jian Yang, Jiaheng Liu, Xianjie Wu,	Kangning Zhang, Wenxiang Jiao, Kounianhua Du,	1798
1745	Changzai Pan, Jie Zhang, Yu Zhao, Shuangyong Song, Yongxiang Li, , and Zhoujun Li. 2025c. Tabler1: Region-based reinforcement learning for table understanding. ArXiv preprint arXiv:2505.12415.	Yuan Lu, Weiwen Liu, Weinan Zhang, , and Yong Yu. 2025b. Looptool: Closing the data-training loop for robust llm tool calls. ArXiv preprint arXiv:2511.09148.	1799
1746			1800
1747			1801
1748			1802
1749			
1750	Can Xie, Ruotong Pan, Xiangyu Wu, Yunfei Zhang,	Ruipeng Zhang, Ya-Chien Chang, , and Sicun Gao. 2025c. When maximum entropy misleads policy optimization. ArXiv preprint arXiv:2506.05615.	1803
1751	Jiayi Fu, Tingting Gao, , and Guorui Zhou. 2025. Unlocking exploration in rlvr: Uncertainty-aware advantage shaping for deeper reasoning. ArXiv preprint arXiv:2510.10649.		1804
1752			1805
1753		Shengjia Zhang, Weiqin Yang, Jiawei Chen, Peng Wu, Yuegang Sun, Gang Wang, Qihao Shi, , and Can Wang. 2026a. Talos: Optimizing top-k accuracy in recommender systems. ArXiv preprint arXiv:2601.19276.	1806
1754			1807
1755	An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao,		1808
1756	Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, , et al. 2024a. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. ArXiv preprint arXiv:2409.12122.		1809
1757			1810
1758		Wenyuan Zhang, Xinghua Zhang, Haiyang Yu, Shuaiyi Nie, Bingli Wu, Juwei Yue, Tingwen Liu, , and Yongbin Li. 2026b. Expseek: Self-triggered experience seeking for web agents. Preprint, arXiv:2601.08605.	1811
1759			1812
1760			1813
1761	Shihui Yang, Chengfeng Dou, Peidong Guo, Kai Lu, Qiang Ju, Fei Deng, , and Rihui Xin. 2025a. Dcpo: Dynamic clipping policy optimization. ArXiv preprint arXiv:2509.02333.		1814
1762		Xingjian Zhang, Siwei Wen, Wenjun Wu, , and Lei Huang. 2025d. Edge-grpo: Entropy-driven grpo with guided error correction for advantage diversity. ArXiv preprint arXiv:2507.21848.	1815
1763			1816
1764			1817
1765	Weiqin Yang, Jiawei Chen, Xin Xin, Sheng Zhou, Binbin Hu, Yan Feng, Chun Chen, , and Can Wang. 2024b. Psl: Rethinking and improving softmax loss from pairwise perspective for recommendation. advances in neural information processing systems, 37:120974–121006.		1818
1766		Xinglang Zhang, Yunyao Zhang, ZeLiang Chen, Junqing Yu, Wei Yang, , and Zikai Song. 2026c. Logical phase transitions: Understanding collapse in llm logical reasoning. Preprint, arXiv:2601.02902.	1819
1767			1820
1768			1821
1769			1822
1770			
1771	Weiqin Yang, Jiawei Chen, Shengjia Zhang, Peng Wu, Yuegang Sun, Yan Feng, Chun Chen, , and Can Wang. 2025b. Breaking the top-k barrier: Advancing top-k ranking metrics optimization in recommender systems. In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2, pages 3542–3552.	Yunyao Zhang, Xinglang Zhang, Junxi Sheng, Wenbing Li, Junqing Yu, Yi-Ping Phoebe Chen, Wei Yang, , and Zikai Song. 2026d. Semantic-aware logical reasoning via a semiotic framework. Preprint, arXiv:2509.24765.	1823
1772			1824
1773			1825
1774			1826
1775			1827
1776		Ziqi Zhao, Zhaochun Ren, Jiahong Zou, Liu Yang, Zhiwei Xu, Xuri Ge, Zhumin Chen, Xinyu Ma, Daiting Shi, Shuaiqiang Wang, , et al. 2026. Reinforced efficient reasoning via semantically diverse exploration. ArXiv preprint arXiv:2601.05053.	1828
1777			1829
1778	Weiqin Yang, Bohao Wang, Zhenxiang Xu, Jiawei Chen, Shengjia Zhang, Jingbang Chen, Canghong Jin, , and Can Wang. 2026. Bear: Towards beam-searchaware optimization for recommendation with large language models. ArXiv preprint arXiv:2601.22925.		1830
1779			1831
1780			1832
1781			
1782			
1783	Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, , et al. 2025. Dapo: An open-source llm reinforcement learning system at scale. ArXiv preprint arXiv:2503.14476.	Yixiao Zhou, Yang Li, Dongzhou Cheng, Hehe Fan, , and Yu Cheng. 2026. Look inward to explore outward: Learning temperature policy from llm internal states via hierarchical rl. ArXiv preprint arXiv:2602.13035.	1833
1784			1834
1785			1835
1786			1836
1787			1837
1788	Weihaio Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, , and Junxian He. 2025. Simplerlzo: Investigating and taming zero reinforcement learning for open base models in the wild. ArXiv preprint arXiv:2503.18892.	Xinyu Zhu, Mengzhou Xia, Zhepei Wei, Wei-Lin Chen, Danqi Chen, , and Yu Meng. 2025. The surprising effectiveness of negative reinforcement in llm reasoning. ArXiv preprint arXiv:2506.01347.	1838
1789			1839
1790			1840
1791			1841
1792			
1793	Kaiyan Zhang, Yuxin Zuo, Bingxiang He, Youbang Sun, Runze Liu, Che Jiang, Yuchen Fan, Kai Tian, Guoli Jia, Pengfei Li, , et al. 2025a. A survey of	Brian D Ziebart, Andrew Maas, J Andrew Bagnell, , and Anind K Dey. 2008. Maximum entropy inverse reinforcement learning. In AAAI, volume 8, pages 1433–1438.	1842
1794			1843
1795			1844
			1845

A Closed Forms for Section 4.4

Perturbing one logit of a k -way uniform distribution by δ and writing $S := e^\delta + k - 1$ gives $\pi_1 = e^\delta/S$ and $\pi_j = 1/S$ for $j \neq 1$, so

$$H(\delta) = \log S - \frac{\delta e^\delta}{S},$$

$$H'(\delta) = -\frac{(k-1)\delta e^\delta}{S^2},$$

whence $H'(0) = 0$ and $H''(0) = -(k-1)/k^2$. The sign of H' is $-\text{sign}(\delta)$, so any perturbation lowers the entropy, but only at second order. Table 13 contrasts this with a peaked distribution, where the first-order term does not vanish.

distribution	H_0	Ω	ΔH at $\delta=.1$
uniform, $k=10$	2.303	0	-4.7×10^{-4}
uniform, $k=50$	3.912	0	-1.0×10^{-4}
top .5 + 9 uniform	1.792	-0.549	-5.6×10^{-2}
top .9 + 9 uniform	0.545	-0.396	-3.8×10^{-2}

Table 13: First-order prediction versus actual entropy change. At a uniform branch point the first-order term is exactly zero and the realized damage is two orders of magnitude below that of a peaked state under the same perturbation.

B Implementation Notes

Sibling mask. For a group of G rollouts of length T we form $\text{eq}[i, j, t] = \mathbf{1}[y_t^{(i)} = y_t^{(j)}]$ and take a cumulative product along t , so that $\text{prefixeq}[i, j, t]$ is true exactly when the two rollouts agree on $y_{\leq t}$. Shifting it one position right gives agreement on $y_{< t}$, which is the sibling relation Eq. (15) requires. Padded positions are treated as disagreement, so a length difference counts as a divergence.

Alignment. $\hat{H}_{\text{togo}}$ is read at position $t+1$ and moved to position t ; at the final position the value is kept. Omitting this shift scores the state before the branch instead of the branch taken.

Memory. The RL path never materializes head logits. For each head the hidden states are processed in chunks of 4096 rows, mapped through the shared unembedding, reduced to an entropy, and released, so peak memory is $O(\text{chunk} \times V)$ rather than $O(\kappa \times B \times T \times V)$.

Equivalence test. A unit test asserts that with $\lambda = 0$ the token weights are bitwise identical to the base method’s on random inputs; the $\lambda = 0$ early

return in the combination step is what makes this hold exactly rather than to floating-point tolerance.

C Pre-Registered Analysis

The analysis plan below was committed to the artifact repository before any run of the multi-seed campaign was launched, so that the statistics could not be chosen to fit the outcome. It is reproduced here in full; the file is `docs/preregistration_5seed.md` in the release.

Unit and primary endpoint. The unit of analysis is the seed, not the validation step. Each run is reduced to one number — its mean `acc@32` on AIME24 over steps 40–110 — and arms are paired within a seed.

Contrasts, in the order they are reported. Primary: STEER-F – GRPO. Secondary, in this order: STEER-F – STEER, STEER-F – UNIFORM, STEER-F – PERMUTED. Secondary endpoints: `maj@32`, uplift, and the six-benchmark mean. Effect sizes are reported with intervals rather than significance stars, and all five arms are reported whatever the outcome.

Stopping and exclusion. Every run goes to step 110; none is stopped early for looking good or bad. A run is excluded only for mechanical failure — out of memory, a crash, a corrupted checkpoint, or a step-0 validation score outside .030–.055, which would indicate a different initial checkpoint. Exclusions are listed with their reason. A failed run is relaunched with the same seed, never a new one, and no seed is added after results are seen.

Checkpoint selection. Two rules are computed and both reported: the argmax over a run’s validation points, and a fixed step 110. The six-benchmark evaluation uses the argmax. Selecting a maximum over twelve noisy checkpoints is biased upward — in the single-seed data by +.011 to +.017 per arm — which is why the primary endpoint is a window mean rather than a maximum.

What would falsify the central claim. Stated in advance so it cannot be renegotiated afterwards. If STEER-F – UNIFORM straddles zero across seeds, the forecast’s *value* does not matter and the apparatus explains the effect; that is the claim of this paper and it would be refuted. If STEER-F – PERMUTED straddles zero, the pairing between a branch and its score does not matter, which the derivation says

it must. If aggregate entropy orders the arms the same way accuracy does, the dissociation argument of Section 12.5 fails, since it depends on the two orderings coming apart. And if the mean token weight in the STEER-F arm departs from ≈ 0.999 , then A_H is not sum-zero over sibling sets as derived and the implementation is wrong rather than the theory.

Amendment, recorded. The pre-registration assigned three seeds to UNIFORM and PERMUTED and five to the other three arms. The campaign as executed runs all five arms at five seeds. The change raises the power of the two control contrasts and was made before any result of the campaign was seen; it is recorded here rather than silently absorbed.

D Within-Run Stability

Earlier drafts of this work paired the arms at each of the eight validation steps in the converged window and reported a paired t over those eight points. That statistic answers a narrower question than the one in Section 12.3: it asks whether a gap is stable along a single trajectory, not whether it reappears when the run is repeated. Its n counts validation points, which are serially correlated within a run, so it is not an estimate of run-to-run uncertainty and cannot be read as one.

We report it here because it is informative about a different thing — whether a gap is an artifact of one lucky validation point — and because the single-seed version of this manuscript quoted it. Where the two disagree, the seed-level statistic in Section 12.8 governs.

The contrast between the two is the point. Table 14 separates every A_H comparison at t_7 between +2.56 and +7.18, which reads as decisive and is not: the same gap sits at the same order as the run-to-run spread we measure in Section 12.8. A statistic can be this confident about a trajectory and say nothing about a re-run, which is exactly why we moved it out of the main text.

contrast	$\Delta\text{acc@32}$	paired t_7
STEER-F – GRPO	+0.0145	+3.08
STEER-F – STEER	+0.0165	+6.68
STEER-F – UNIFORM	+0.0182	+7.18
STEER-F – PERMUTED	+0.0125	+2.56

Table 14: Within-run stability over the eight converged-window validation points of seed 1. The points are consecutive validations of one run and are therefore serially correlated, so $\text{df} = 7$ is generous and these t values are not an estimate of run-to-run variation; see Section 12.8 for that. Majority-vote versions of the same four rows separate at +5.09 to +5.50.